Chapter: Introduction

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Exercise 1.8

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

By using a change of variable, verify that the univariate Gaussian distribution given by (1.46) statisfies (1.49).
let restate the equations (1.46) and (1.49) for clarity:

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\} \tag{1.46}$$

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2) x dx = \mu \tag{1.49}$$

To verify that the univariate Gaussian distribution satisfies the expectation value given by (1.49), we can perform a change of variable. Let's define a new variable z as follows:

$$z = x - \mu \implies x = z + \mu$$

Now, we can rewrite the expectation value in terms of z:

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2) x dx = \int_{-\infty}^{\infty} \mathcal{N}(z+\mu|\mu,\sigma^2)(z+\mu) dz$$

Substituting $x = z + \mu$ and the expression for the Gaussian distribution, we get:

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\}(z+\mu) dz$$

We can now split this integral into two parts and factor out the constants:

$$\begin{aligned}\mathbb{E}[x] &= \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z dz + \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} \mu dz \\ &= \frac{1}{(2\pi\sigma^2)^{1/2}} \left[\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z dz + \mu \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} dz\right]\end{aligned}$$

By refering to exercise 1.7, we know that

$$I = \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} dz = (2\pi\sigma^2)^{1/2} \quad \text{with any abitrary z term}$$

By rewriting $\mathbb{E}[x]$ with the value of I, we get:

$$\begin{aligned}\mathbb{E}[x] &= \frac{1}{(2\pi\sigma^2)^{1/2}} \left[\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z dz + \mu(2\pi\sigma^2)^{1/2}\right] \\ &= \mu + \frac{1}{(2\pi\sigma^2)^{1/2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z dz\end{aligned}$$

Let us denote the remaining integral as J and prove that it is equal to zero:

$$J = \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z dz$$

Let show that $\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z$ is an odd function, and thus its integral over the entire real line is zero.
To show that the function $f(z) = \exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z$ is an odd function, we need to verify that $f(-z) = -f(z)$.
Let's compute $f(-z)$:

$$f(-z) = \exp\left\{-\frac{1}{2\sigma^2}(-z)^2\right\}(-z) = \exp\left\{-\frac{1}{2\sigma^2}z^2\right\}(-z) = -\exp\left\{-\frac{1}{2\sigma^2}z^2\right\} z = -f(z)$$

Since $f(-z) = -f(z)$, the function $f(z)$ is indeed an odd function. Therefore, the integral of $f(z)$ over the entire real line is zero:

$$J = 0$$

By substituting $J = 0$ back into the expression for $\mathbb{E}[x]$, we get:

$$\mathbb{E}[x] = \mu + \frac{1}{(2\pi\sigma^2)^{1/2}} \cdot 0 = \mu$$

Thus, we have verified that the univariate Gaussian distribution satisfies the expectation value given by (1.49), which is $\mathbb{E}[x] = \mu$.

By differentiating both sides of the normalization condition with respect to σ^2, verify that the univariate Gaussian distribution given by (1.46) satisfies (1.50).
Let's restate the normalization condition and equation (1.50) for clarity:

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2)x^2 dx = \mu^2 + \sigma^2 \tag{1.50}$$

To verify that the univariate Gaussian distribution satisfies the expectation value given by (1.50), we can differentiate both sides of the normalization condition with respect to σ^2.
The normalization condition states that the integral of the Gaussian distribution over the entire real line is equal to 1:

$$\int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2) dx = 1$$

Differentiating both sides of this equation with respect to σ^2, we get:

$$\frac{d}{d\sigma^2}\int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2) dx = \frac{d}{d\sigma^2}1$$

Since the right-hand side is a constant, its derivative is zero:

$$\int_{-\infty}^{\infty} \frac{\partial}{\partial\sigma^2}\mathcal{N}(x|\mu,\sigma^2) dx = 0$$

Now, we need to compute the partial derivative of the Gaussian distribution with respect to σ^2:

$$
\begin{aligned}
\frac{\partial}{\partial\sigma^2}\mathcal{N}(x|\mu,\sigma^2) &= \frac{\partial}{\partial\sigma^2}\left(\frac{1}{(2\pi\sigma^2)^{1/2}}\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}\right)\\
&= \frac{\partial}{\partial\sigma^2}\left(\frac{1}{(2\pi\sigma^2)^{1/2}}\right)\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\} + \frac{1}{(2\pi\sigma^2)^{1/2}}\frac{\partial}{\partial\sigma^2}\left(\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}\right)\\
&= -\frac{1}{(\pi)^{1/2}(2\sigma^2)^{3/2}}\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\} + \frac{1}{(2\pi\sigma^2)^{1/2}}\left(\frac{1}{2\sigma^4}(x-\mu)^2\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}\right)\\
&= \frac{1}{(2\pi\sigma^2)^{1/2}}exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}\left(\frac{1}{2\sigma^4}(x-\mu)^2-\frac{1}{2\sigma^2}\right)\\
&= \mathcal{N}(x|\mu,\sigma^2)\left(\frac{1}{2\sigma^4}(x-\mu)^2-\frac{1}{2\sigma^2}\right)
\end{aligned}
$$

Substituting this back into the integral, we get:

$$\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)\left(\frac{1}{2\sigma^4}(x-\mu)^2-\frac{1}{2\sigma^2}\right)dx = 0$$

We can split this integral into two parts:

$$\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)\frac{1}{2\sigma^4}(x-\mu)^2dx - \int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)\frac{1}{2\sigma^2}dx = 0$$

Now, we can factor out the constants:

$$\frac{1}{2\sigma^4}\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)(x-\mu)^2dx - \frac{1}{2\sigma^2}\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)dx = 0$$

We know that the integral of the Gaussian distribution over the entire real line is equal to 1:

$$\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)dx = 1$$

Thus, we can simplify the equation to:

$$
\begin{aligned}
\frac{1}{2\sigma^4}\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)(x-\mu)^2dx - \frac{1}{2\sigma^2} &= 0\\
\frac{1}{2\sigma^4}\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)(x-\mu)^2dx &= \frac{1}{2\sigma^2}\\
\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)(x-\mu)^2dx &= \sigma^2
\end{aligned}
$$

By manipulating $V = \int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)(x-\mu)^2dx$, we can express it in terms of $\mathbb{E}[x^2]$ and $\mathbb{E}[x]$:

$$
\begin{aligned}
V &= \int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)(x^2-2\mu x+\mu^2)dx\\
&= \int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)x^2dx - 2\mu\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)xdx + \mu^2\int_{-\infty}^{\infty}\mathcal{N}(x|\mu,\sigma^2)dx\\
&= \mathbb{E}[x^2] - 2\mu\mathbb{E}[x] + \mu^2 \quad (\text{since } \mathbb{E}[x] = \int_{-\infty}^{\infty}x\mathcal{N}(x|\mu,\sigma^2)dx \text{ and } \mathbb{E}[x^2] = \int_{-\infty}^{\infty}x^2\mathcal{N}(x|\mu,\sigma^2)dx)\\
&= \mathbb{E}[x^2] - 2\mu^2 + \mu^2 \quad (\text{since } \mathbb{E}[x] = \mu)\\
&= \mathbb{E}[x^2] - \mu^2
\end{aligned}
$$

Since we have already established that $V = \sigma^2$, we can substitute this back into the equation to find $\mathbb{E}[x^2]$:

$$\begin{aligned} \sigma^2 &= \mathbb{E}[x^2] - \mu^2 \\ \mathbb{E}[x^2] &= \mu^2 + \sigma^2 \end{aligned}$$

Thus, we have verified that the univariate Gaussian distribution satisfies the expectation value given by (1.50), which is $\mathbb{E}[x^2] = \mu^2 + \sigma^2$.

Finally let show (1.51) by using the results of (1.49) and (1.50):

$$\begin{aligned} \mathrm{var}[x] &= \mathbb{E}[x^2] - (\mathbb{E}[x])^2 \\ &= (\mu^2 + \sigma^2) - \mu^2 \\ &= \sigma^2 \end{aligned}$$